

Esteban Sitek

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with **12+** years across fintech, healthcare, and e-commerce, building production ETL pipelines and analytics foundations using Python, PySpark, and dbt. Skilled in data modeling, quality assurance, and applied data science with Streamlit and predictive analytics. Proven track record delivering pragmatic solutions that empower business stakeholders with actionable insights.

SKILLS

- Languages:** Python, SQL, JavaScript, TypeScript, HTML, CSS
- Data & ML:** PySpark, Polars, dbt, Airflow, Pandas, NumPy, Spark, scikit-learn, TensorFlow, MLflow
- Backend:** FastAPI, Node.js, Express, GraphQL, gRPC
- Frontend:** React, Streamlit, Next.js, Redux, Tailwind CSS
- Database:** PostgreSQL, DuckDB, MongoDB, Redis, Elasticsearch, MySQL
- Testing:** Pytest, Jest, Playwright, Cypress, unittest
- Cloud & DevOps:** Docker, Kubernetes, Git, GitHub, AWS, GCP, GitHub Actions, Jenkins, Terraform
- Others:** RESTful API, Kafka, RabbitMQ, Data modeling, ETL pipelines, Business analytics, Jira, Confluence

EXPERIENCE

Publicis Sapient

Senior Software Engineer

Jun 2024–Present

- Led development of data integration workflows for enterprise analytics platforms, architecting **Python** and **FastAPI** microservices that ingested and transformed data from multiple third-party SaaS vendors into a unified reporting layer.
- Collaborated with analytics teams to design and document data schemas using **SQL** and **PostgreSQL**, ensuring consistency across 12+ analytics domains and reducing report generation time by 40% through optimized query patterns.
- Drove adoption of **dbt** for data transformation logic, establishing version control and testing standards that improved data quality scores from 82% to 98% and enabled non-technical stakeholders to validate transformations through pre-built tests.
- Implemented monitoring and alerting using **Prometheus** and **Grafana** to track pipeline health across a distributed system serving 500k+ daily data points, catching and resolving latency issues before they impacted downstream analytics.

- Mentored two junior engineers on **Docker** and **Kubernetes** deployment practices, establishing CI/CD workflows using **GitHub Actions** that reduced manual deployment overhead and incident response time by 35%.
- Built **Streamlit** dashboards for stakeholder-facing analytics, translating complex product datasets into interactive visualizations that reduced weekly report generation from 8 hours to 1 hour.
- Evaluated and integrated **DuckDB** as an alternative to traditional data warehouses for analytical queries, achieving **3x** faster query latency on medium-scale datasets while reducing infrastructure costs by 25%.
- Participated in R&D initiatives to prototype **TensorFlow**-based demand forecasting models, delivering a proof-of-concept that improved inventory predictions by 18% with minimal computational overhead.
- Conducted code reviews and pair programming sessions focused on data quality validation, helping the team identify and prevent 23 critical data inconsistencies before reaching production.
- Documented data lineage and ETL logic across 8 critical pipelines using internal wikis and **Confluence**, enabling faster onboarding and reducing troubleshooting time for on-call incidents by 30%.
- Worked closely with product and business teams to gather analytics requirements, translating ambiguous stakeholder requests into concrete data models and reducing scope creep by 45%.

CareSimple

Oct 2022–May 2024

Senior Full Stack Engineer

- Spearheaded the rebuild of the patient telemetry data pipeline using **PySpark** and **Airflow**, replacing a legacy batch job that had been losing 2-3% of critical health records daily with a fault-tolerant streaming architecture that guaranteed 99.99% delivery.
- Owned the design and implementation of data quality frameworks using **Pandas** and **Pytest**, establishing automated validation rules for 40+ clinical metrics that caught malformed patient records before they reached downstream analysis systems.
- Collaborated with data scientists to integrate **scikit-learn** models into the population health platform, operationalizing risk prediction algorithms that reduced hospital readmission rates by 12% across 18 health systems.
- Built internal analytics APIs using **FastAPI** and **PostgreSQL**, enabling clinical teams to query 3 years of de-identified patient outcomes in seconds rather than hours and supporting 15+ concurrent analytical users.
- Designed and documented the telemetry data schema to support longitudinal analysis, modeling patient journeys across multiple encounters using **SQL** window functions that enabled clinical researchers to identify treatment patterns previously buried in unstructured notes.
- Implemented **Redis**-backed caching layer for frequently accessed patient cohorts, reducing dashboard load times from 45 seconds to 3 seconds and improving clinician adoption of analytics by 60%.
- Established data governance policies and audit logging using **Kafka** event streams, ensuring HIPAA compliance and enabling forensic analysis of data access patterns across 12 healthcare organizations.

Shopee

Jul 2018–Sep 2022

Full Stack Engineer

- Built scalable ETL pipelines processing 50 million daily transactions using **Spark** and **Airflow**, modeling merchant performance data and consumer behavior patterns to feed A/B testing and recommendation engine teams.
- Designed the product catalog data model in **PostgreSQL** and **DuckDB**, creating a star schema that reduced analytics query latency from 8 minutes to 12 seconds and enabled real-time inventory dashboarding across 300k+ sellers.
- Developed data validation and reconciliation processes using **Python** and **Pandas**, detecting inventory sync issues between warehouses and marketplaces within minutes and preventing \$2.3 million in customer refunds annually.
- Collaborated with the monetization team to instrument **Kafka** event tracking across the payment funnel, capturing 15+ conversion metrics that fed into **Streamlit** dashboards used by product managers to optimize checkout flows and increase transaction completion by 8%.
- Implemented feature engineering pipelines using **scikit-learn** and **MLflow**, supporting the recommendations team's model training and A/B testing infrastructure and reducing time-to-experiment from 3 weeks to 5 days.
- Mentored 3 junior engineers on **Git** workflows, **Docker** containerization, and **Kubernetes** deployment, establishing standards that reduced production incidents related to deployment from 12 per month to 2 per month.
- Owned the fraud detection data platform, ingesting transaction logs, device fingerprints, and merchant metadata into **MongoDB** and building **Elasticsearch**-backed search APIs that allowed fraud analysts to investigate suspicious patterns in real-time.

Altitude Games

Nov 2014–Jul 2018

Full Stack Engineer

- Built analytics ingestion pipelines for mobile game events using **Python** and **SQL**, capturing 200k+ daily player events and modeling progression, retention, and monetization metrics that informed game balance decisions.
- Designed and maintained the player progression database schema in **PostgreSQL**, optimizing queries to support real-time leaderboards serving 500k concurrent players and reducing query latency from 800ms to 150ms.
- Developed backend services using **Node.js** and **Express** to support in-game economy systems, processing merchant transactions and player-to-player trades with **Redis**-backed eventual consistency that enabled concurrent operations without inventory conflicts.
- Contributed to data analysis work using **Pandas** and statistical methods to identify player cohorts and predict churn, supporting retention initiatives that improved 30-day retention by 7% across the player base.
- Implemented **Kafka** event streaming to capture gameplay telemetry from mobile clients, persisting 60 million daily events to **MongoDB** and enabling retrospective analysis of player behavior patterns.

EDUCATION

Ateneo de Manila University
Bachelor of Science in Computer Science

Aug 2010–Jul 2014